

SOC ALERT SIMULATION REPORT

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1. Incident Overview

During monitoring of system security logs, multiple failed login attempts were detected on a Windows machine, indicating a potential unauthorized access attempt. These repeated failures suggest a potential unauthorized access attempt targeting the system.

2. Objective

The objective was to simulate and detect suspicious login activity by analyzing security events and understanding how SOC alerts are generated. This includes analyzing Windows Security Logs to understand how Security Operations Center (SOC) alerts are triggered and investigated.

3. Methodology

A controlled simulation was performed by generating multiple failed login attempts on a Windows system. The logs were monitored and analyzed using Windows Event Viewer under Security logs. Relevant events were filtered and analyzed for suspicious patterns.

4. Observations

- A total of 6 failed login attempts were recorded.
- All events correspond to Event ID 4625 (Failed Login/Logon).
- The events are marked as Audit Failure.
- Multiple attempts occurred within a short time interval.
- The system account involved was SYSTEM / MSI\$.
- Status Codes observed: 0xC000006D (Bad credentials) and 0xC0000380 (Authentication failure detail).

5. Identified Issue

The repeated failed login attempts indicate a **brute force attack pattern**, where multiple incorrect passwords or combinations are used to try to gain unauthorized access to the system.

6. Affected System

System Type:	Windows Machine
Computer Name:	MSI
Log Source:	Microsoft Windows Security Auditing

7. Evidence

- Event ID: 4625.
- Event Type: Audit Failure.
- Number of Attempts: 6.
- Log Message: "An account failed to log on".
- Timestamp: Multiple entries within a short duration/interval.

8. Impact Analysis

- Increased risk of unauthorized system access.
- Potential account compromise if the attack succeeds.
- Indicates vulnerability to password-based attacks (Brute Force).
- Indicates weak authentication or lack of protection mechanisms.

9. Response Strategy

To mitigate such threats, the following actions are recommended:

- Temporarily lock the affected account or enable Account Lockout Policy.
- Enforce strong password policies.
- Enable/Implement Multi-Factor Authentication (MFA).
- Monitor login activity continuously using SIEM tools for real-time alerts.
- Block suspicious IP addresses and conduct regular security audits.

10. Conclusion

The analysis confirms a simulated brute force attack scenario based on repeated failed login attempts identified through Event ID 4625. This highlights how SOC analysts detect and respond to suspicious authentication activity to prevent security breaches.

